

Particle motion in a fluid

The so-called Maxey-Riley-Gatignol equations (MRGE)

$$y'(t) = v(t) \quad (1)$$

$$Rv'(t) = \frac{Du}{Dt}(t) - \frac{1}{S}(v(t) - u(t)) - \sqrt{\frac{3}{\pi S}} \left\{ \frac{v(0) - u(0)}{\sqrt{t}} + \int_0^t \frac{v'(s) - u'(s)}{\sqrt{t-s}} ds \right\} \quad (2)$$

describe the motion of particles in a fluid, provided that they do not alter the fluid flow. This is an appropriate assumption when the particles are far smaller than the relevant fluid flow scales, and when the suspension of particles is dilute, see [1]. For example, the MRGE are used for simulating the settling of plankton in the ocean (“marine snow”) [2, 3]. Hereby $u(y(t), t)$ is the fluid velocity at the particle position $y(t)$ and $v(t) = y'(t)$ is the particle velocity. The term $\frac{Du}{Dt} := \frac{\partial u}{\partial t} + v \cdot \nabla u$ with $[v \cdot \nabla u]_i := \sum_j \frac{\partial u_i}{\partial y_j} v_j$ is the so-called material derivative.¹ The parameters are

$$R := \frac{1 + 2\rho_p/\rho_f}{3} \quad \text{and} \quad S := \frac{a^2}{3T\nu}, \quad (3)$$

where ρ_p and ρ_f are particle and fluid density, a is the particle diameter, ν is the fluid viscosity, and T is the time scale associated with the flow. The integral term in (2) is called history or Basset term. We neglect this difficult term and take the simplified equation

$$Rv'(t) = \frac{Du}{Dt}(t) - \frac{1}{S}(v(t) - u(t)). \quad (4)$$

We want to simulate the motion of a single particle in a two-dimensional vortex flow, cf. [1], Eq. (37):

$$u(t, y) = u(y) := \begin{pmatrix} -y_2 \\ y_1 \end{pmatrix}. \quad (5)$$

Your Tasks are as follows:

- I) Compute the material derivative $\frac{Du}{Dt}$ for the vortex flow (5) and write down the ODE system $w' = f(t, w)$, where $w := \begin{pmatrix} y \\ v \end{pmatrix}$. Then, compute the analytic solution w for given initial value vector $w_0 := w(0)$. **Hint:** Use the matrix exponential.²
- II) Solve the ODE for $R := 7/9$, $S := 0.01$, initial values $y(0) := (1, 0)^\top$, $v(0) := (0, 1)^\top$, and $t \in [0, 10]$ by different methods of Python’s ODE solver `solve_ivp`³. For example, try `RK45`, `RADAU`, `BDF`, or others. Compare the results, the step sizes, and the number of steps used by each method. What do you observe? What can be said about stiffness of the problem in dependence on the parameters R and S ? Which method do you recommend? Give concise reasons for your answers. Especially Chapter 6 of the script might be helpful.
- III) Create two phase portraits visualizing the particle trajectory $y(t) = (y_1(t), y_2(t))$ for the following two cases $S := 0.3$, $R := 7/9 < 1$ (particle lighter than fluid) and $S := 0.3$, $R := 4/3 > 1$ (particle heavier than fluid). Initial values and integration period are the same as in I). Compare your phase portraits with those in [1], Fig. 1. Based on that, judge the influence of the neglected history term in (2).

¹https://en.wikipedia.org/wiki/Material_derivative

²https://en.wikipedia.org/wiki/Matrix_exponential

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.linalg.expm.html>

³https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html

IV) Implement the following functions in YAPS:

- a) `runge_kutta(f, Jf, y0, t, a, B, c)`, general explicit or diagonally implicit Runge-Kutta method. For the latter, you must implement a Newton iteration for solving the arising nonlinear systems of equations. Stop the iteration if the norm of the difference of the iterates is $\leq 1e-8$.
- b) `k_step(f, Jf, y0, y1, t, a, b)`, general explicit or implicit k-step method. As in Project 2 you can use Python's built-in ODE solvers like `odeint` for generating missing starting values. For implicit methods you must use a Newton iteration as in a).
- c) `f(t, w, R, S)`, right-hand side of the ODE $w' = f(t, w, R, S)$ with parameters R and S .
- d) `Jf(t, w, R, S)`, Jacobian of f with respect to w (needed for Newton iteration).
- e) `w_exact(t, R, S, w0)`, analytical solution of $w' = f(t, w, R, S)$, $w(0) = w_0$ as computed in I).

For all five functions, templates with parameter description and examples are given in the file `exercise.py` which you can download in StudIP/ILIAS. You shall complete these templates. Before you submit your functions in YAPS, you can a priori check them offline with the main program `__main__.py` which is also available in StudIP/ILIAS. (Finally, YAPS will run additional tests of your code which we do not disclose.)

Submission: Your submission consists of two parts

1. A technical report of up to 2 pages as PDF submitted to ILIAS, see StudIP for the link.
File name: `LASTNAME.GIVENNAME.pdf`
2. A Python code `exercise.py` verified in YAPS.
The YAPS system will be open from **29.06. to 07.07.**
Link: <https://yaps.amh.tuhh.de/numode/project-3-ss26-new/examserver/client/instance/>

Deadline: 07.07.2026.

References

- [1] J. Urizarna-Carasa, L. Schlegel, D. Ruprecht, Efficient numerical methods for the Maxey-Riley-Gatignol equations with Basset history term, *Computer Physics Communications*, Vol. 310, 109502, 2025. <https://arxiv.org/pdf/2403.13515>
- [2] M. Niazi Ardekani, G. Sardina, L. Brandt, L. Karp-Boss, R. Bearon, e. Variano, Sedimentation of elongated non-motile prolate spheroids in homogenous isotropic turbulence, *arXiv:e-prints*, arXiv-1611, 2016. <https://arxiv.org/pdf/1611.10284>
- [3] K. Guseva, A. Daitche, U. Feudel, T. Tél, History effects in the sedimentation of light aerosols in turbulence: the case of marine snow, *Phys. Rev. Fluids* 1 (2016) 074203.
<https://arxiv.org/pdf/1601.01029>

Some general tips for writing reports.

- Do not mix answers to different questions. Every question deserves its own answer that is clearly separate from the others. Do not let the reader figure out which question you are addressing in a given part of the text.
- Every figure needs to be referenced and discussed in the text. Each figure should have a clear “message” associated with it which should be clearly spelled out in the text. If you can’t say what the message of a figure is, you either need to think more about it before writing it down or the figure doesn’t really have one - remove it, then.
Make sure that figures are readable, also when printed in black and white. Whenever possible, use vector formats like PDF for graphs as they scale without loss.
Resist the urge to pad your report with many figures that do not contribute to the key points its making. It will not improve its quality and just look cluttered.
- Keep sentences short. At least one round of proof-reading should be dedicated to identify overly long sentences and to breaking them down.
- At least one additional round of proof-reading should be used to identify and delete waffle. Say what you want and need to say, not more, not less. Being concise is a virtue!
- Use paragraphs.